Behavioral Predictors of Weight Change Among Adults Attempting Weight Loss

Executive Summary (NHANES 2021–2023)

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1 Research question

Which behavioral characteristics best predict 1-year percentage weight change among U.S. adults who tried to lose weight, *beyond* demographics and starting BMI? The outcome is

$$\%\text{ weight change} = \frac{\text{current weight} - \text{weight 1 year ago}}{\text{weight 1 year ago}} \times 100,$$

computed from self-reported weights. Negative values mean weight was lost.

2 Data and sample

- **Source.** Nine public-use NHANES August 2021–August 2023 files (demographics, weight history, physical activity, diet behavior, sleep, smoking, alcohol, depression screener, body measures) were merged on respondent ID, giving 11,933 participants.
- **Sample.** 2,794 adults aged 20 and older who reported trying to lose weight in the past 12 months and attended the MEC exam. Pregnant women were excluded, as were 6 people with changes beyond ±50%. Weighted to the U.S. population, the mean age is 48, 58% are women, and the mean BMI one year earlier was 31.7.
- **Cleaning.** 442 refused or don't-know codes were recoded to missing, column by column, so that genuine answers such as 7 hours of sleep or age 77 were kept. Skip patterns were filled in where the answer was implied; for example, vigorous activity went from 51.5% to 0.5% missing. Measured body-size columns were dropped to avoid outcome leakage.
- **Outcome.** The weighted mean change was −0.9% and the median 0%. About 34% gained weight despite trying to lose it, and 23% reported exactly the same weight as a year earlier.
- **Missing data.** About 12% are missing PHQ-9, drinks per week or income; 76% have no missing values. Missingness was not related to the outcome.

Predictors (27). *Covariates (15):* age, sex, race/ethnicity (5 indicators), education, income-to-poverty ratio, marital status (2 indicators), household size, U.S.-born, veteran status, and BMI one year ago. *Behaviors (12):* moderate and vigorous activity (min/week), sedentary hours/day, depressive symptoms (PHQ-9), drinks/week, average sleep hours, former and current smoking, and four meal-planning and food-shopping roles.

3 Approach

The design compares, for each model, a fit on covariates only with a fit on covariates plus behaviors. The difference in R^2 (ΔR^2) answers the research question.

	Model 1	**Model 2**	**Model 3**
Method	Penalized linear regression: LASSO, Ridge and elastic net	Random forest (1,000 trees)	Gradient boosting (XGBoost)
What it captures	Straight-line, additive effects; LASSO also selects variables	Non-linear effects, thresholds and interactions	Non-linear effects, thresholds and interactions
Missing data	Training-set median imputation + missing indicators for income, PHQ-9 and drinks/week	Handled natively by the model	Handled natively by the model
Tuning	Penalty chosen by 10-fold CV	30-combination randomized search with 5-fold CV	60-combination randomized search with 5-fold CV
Importance and selection	Standardized coefficients, LASSO path, selection stability over 200 bootstrap refits	Permutation importance, SHAP values and interactions; shadow-feature (Boruta-style) selection and refits on smaller predictor sets	Permutation importance (single and grouped), SHAP values and interactions

Shared design. All three models use the same 80/20 train/test split (2,235 / 559), stratified on outcome quintiles. All tuning used the training set only, and the test set was used once. ΔR^2 was assessed by a 2,000-sample paired bootstrap of the test set and by repeated 5-fold cross-validation (25 folds) on the training set.

4 Results

Model performance

Model	Predictors	Test R^2	Test RMSE	Test MAE	CV R^2
Baseline (mean)	—	−0.002	9.09	6.28	—
Model 1: penalized linear regression					
LASSO	Covariates	0.074	8.736	—	0.118
	+ Behaviors	0.080	8.713	—	0.125
Ridge	Covariates	0.072	8.750	—	0.116
	+ Behaviors	0.079	8.715	—	0.123
Elastic net	Covariates	0.075	8.736	—	0.117
	+ Behaviors	0.080	8.713	—	0.125
Model 2: random forest					
Random forest	Covariates	0.071	8.752	6.065	0.122
	+ Behaviors	0.074	8.740	6.093	0.128
Model 3: gradient boosting					
XGBoost	Covariates	0.065	8.783	6.108	0.127
	+ Behaviors	0.075	8.735	6.112	0.138

RMSE and MAE are in percentage points of weight change. CV R^2 is the mean over 25 folds (5-fold CV repeated 5 times). All models beat the mean-only baseline, and the five models perform within about 0.015 of each other on the test set.

Do behaviors add predictive value?

Model	Test ΔR^2 [95% CI]	CV mean ΔR^2	Folds improved
LASSO (Model 1)	0.005 [−0.010, 0.021]	0.008	20 / 25
Ridge (Model 1)	0.008 [−0.011, 0.028]	0.008	20 / 25
Elastic net (Model 1)	0.005 [−0.010, 0.021]	0.008	21 / 25
Random forest (Model 2)	0.003 [−0.015, 0.021]	0.006	19 / 25
XGBoost (Model 3)	0.010 [−0.015, 0.033]	0.011	22 / 25

All five models agree in direction. The gain from behaviors is small and fairly consistent across cross-validation folds, but it cannot be distinguished from zero on the held-out test set alone. The gain is smallest for the random forest.

Which behaviors matter?

Behavior	Direction	**Model 1** (LASSO stability; coef. per SD)	**Model 2** (random forest)	**Model 3** (XGBoost)
Moderate activity	More loss	83%; −0.13	Confirmed by shadow test; test importance clearly above zero; levels off near 300–350 min/wk	Top behavior (mean |SHAP| 0.41); test importance clearly above zero
Vigorous activity	More loss	93%; −0.32	Test importance above zero; rejected by shadow test	Mean |SHAP| 0.24
Main meal planner	More loss	100%; −0.47	Almost no effect	Same direction; small impact
Current smoking	More loss	98%; −0.35	Same direction; small	Same direction
Sedentary time	Less loss	92%; +0.25	Very small effect	Mean |SHAP| 0.28; threshold near 5 h/day
Depressive symptoms (PHQ-9)	Less loss	88%; +0.23	Top behavior by SHAP (0.30) and confirmed, but test importance near zero	Mean |SHAP| 0.33; test importance near zero
Sleep	Short sleep, less loss	82%; weaker sign consistency	Confirmed; U-shape, lowest at 7–8 h; test importance near zero	Mean |SHAP| 0.25; under about 6.5 h linked to less loss
Drinks/week, former smoking, shared meal and shopping roles	—	Kept in about half of resamples	Rejected; almost no effect	Almost no effect

Key findings

- **Starting BMI** is dominant: about −3.0 percentage points per SD (about 7.6 BMI units) in Model 1, roughly six times any other predictor. Shuffling BMI lowers test R^2 by 0.20 in

Model 2 and 0.22 in Model 3, more than either model's entire accuracy.

- **Behaviors as a block** are worth about 0.011 of test R^2 in Model 2 and 0.023 in Model 3. For comparison, the 14 non-BMI covariates are worth 0.034 and 0.043, and BMI alone 0.195 and 0.213.
- **Physical activity is the behavior all three models support.** Model 1 ranks main meal planner first; Models 2 and 3 rank moderate activity first among behaviors with real test-set value. Its benefit levels off after about 150–350 min/week, a curve that a single straight-line slope understates.
- **PHQ-9 and sleep are used but do not generalize.** Both tree models rely on them, yet their test-set permutation importance is about zero. Model 1 keeps PHQ-9 in 88% of resamples, so its role remains uncertain.
- **A compact model works almost as well** (Model 2). A shadow-feature test confirmed six predictors: BMI, age, income-to-poverty ratio, PHQ-9, moderate activity and sleep. A forest on just BMI, age, income-to-poverty ratio and moderate activity reached CV R^2 0.120 and test R^2 0.079, compared with 0.128 and 0.074 for all 27 predictors.
- **Non-linear patterns are descriptive.** The tree models show a plateau for moderate activity, a U-shape for sleep around 7–8 h (Model 2), a sedentary-time threshold near 5 h/day (Model 3), and less weight loss from mild depressive symptoms upward. Because neither tree model predicted better than the linear models, these shapes describe how the models use each behavior rather than proven predictive gains.
- **Interactions are small.** The strongest interactions involve starting BMI (for example, BMI $\times$ moderate activity and BMI $\times$ PHQ-9), but they refine predictions rather than change the overall picture.

5 Conclusion

Among U.S. adults trying to lose weight, starting BMI explains most of what these predictors can say about 1-year percentage weight change. Behavioral measures add a small amount beyond demographics and BMI. Physical activity is the behavioral signal supported by all three model families; meal planning, sedentary time and depressive symptoms are supported mainly by the linear model. A random forest with only four predictors (BMI, age, income-to-poverty ratio and moderate activity) predicts nearly as well as one with all 27. Overall accuracy is modest, which is expected for a self-reported, cross-sectional outcome.